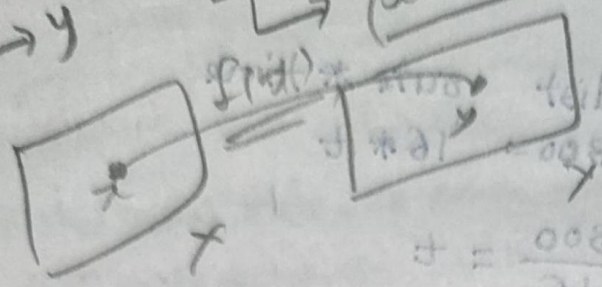
$x \in X$

$f(x) \to y$

$2x+3 \to \boxed{x=5}$



$\boxed{f: X \to Y}$

Vector Transformations

$f: x \to y$

can have multiple vectors

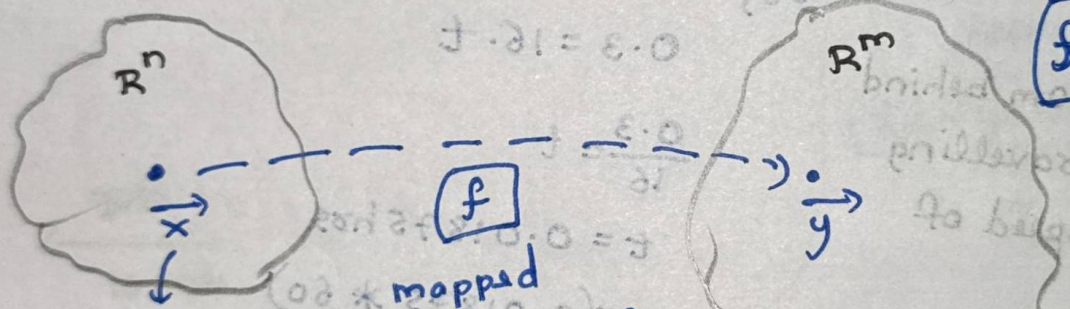
Same as this (inside)

(just a function okay)

$R \to$ Real No.

$\left.\begin{matrix} m \\ n \end{matrix}\right\} \to$ dimensions

$\boxed{f: \mathbb{R}^n \to \mathbb{R}^m}$



(we have a vector here)

mapped because of this fn.

Dimensionality Reduction (Possible)

↳ Say I have a fn. $\to f(x, y, z) = (x+y, 2z)$

$$f\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) \xrightarrow{\text{it becomes}} \begin{bmatrix} 1+2 \\ 2(3) \end{bmatrix} \to \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

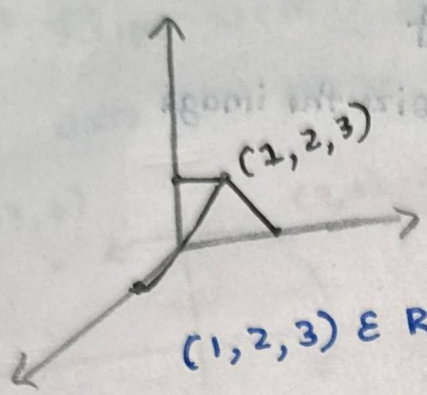
$\to \boxed{x+y}$

↳ 22

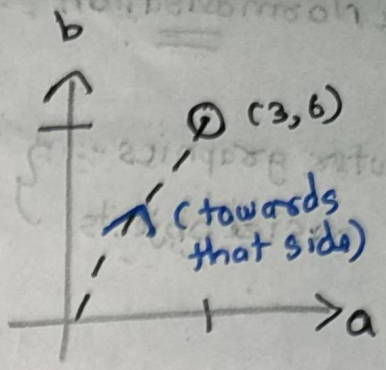
mapping this vector (applying the fn.)

Vector Transformation

Diagrametically



This is getting reduced to $\boxed{f}$



$(1,2,3) \in \mathbb{R}^3 \rightarrow$ 3 dimension

Real values

$\Rightarrow (3,6) \in \mathbb{R}^2 \rightarrow \boxed{2\text{-}D}$

Vector Tranformation

↳ operation that maps vectors from one $\boxed{\text{space}}$ → domain to $\boxed{\text{another}}$ (co-domain)

↳ In this mapping we chang the

① magnitude
② Direction

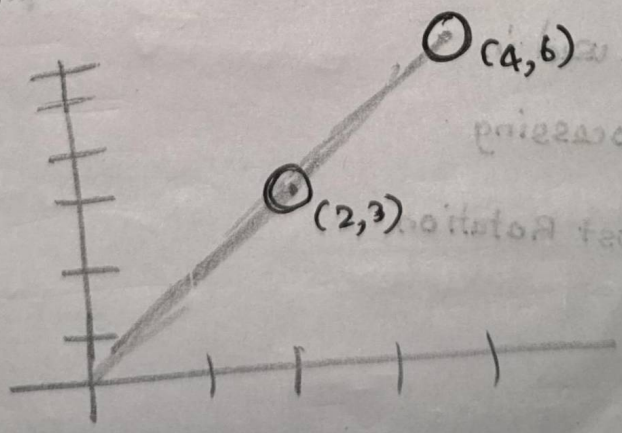
} (or) both → Transformations Described using $\boxed{\text{matrices}}$

Scaling

Direction → Same

magnitude → changes

$$v' = 2v = 2\begin{bmatrix}2\\3\end{bmatrix} \rightarrow \begin{bmatrix}4\\6\end{bmatrix}$$



(Application Part)

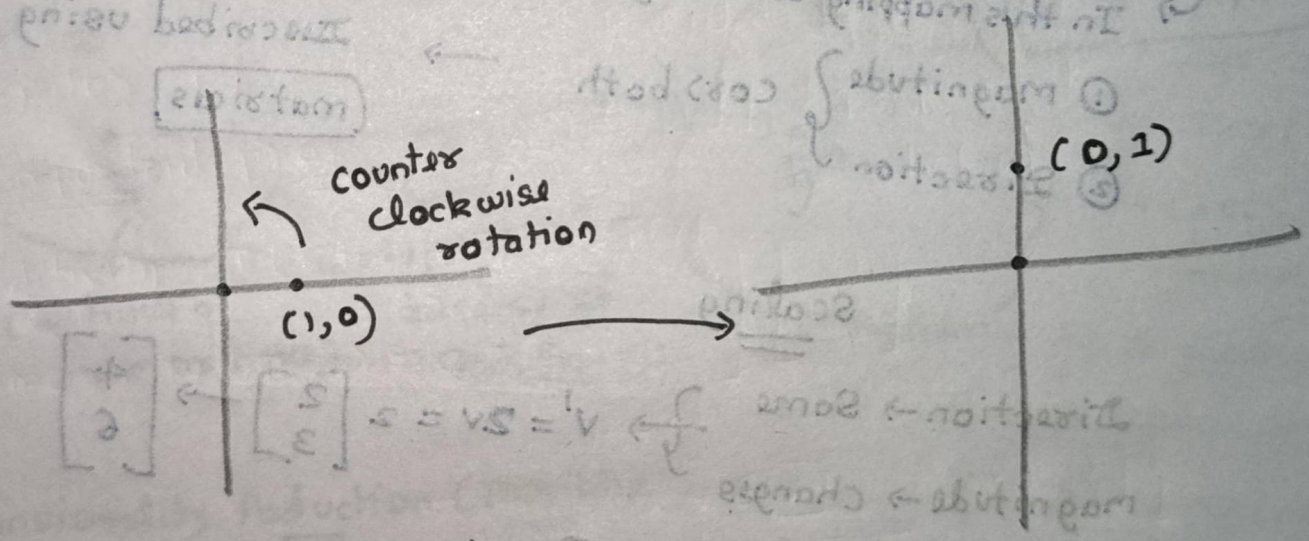
① Data Normalisation

② Computer graphics to resize objects } ⇒ Open Paint and Resize the image

Rotation

Trans that turns vector around the origin

$$V = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \in R^2 \xrightarrow{\text{turns}} V' = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \text{a } 90^\circ \text{ Rotation}$$



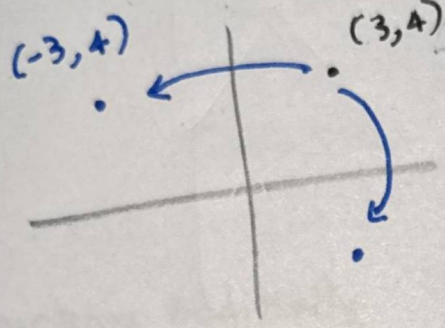
Eg → Rotation will be used in

Image processing

Robotics → Adjust Rotation

③ Reflection

↳ flips vectors over a specified axis (or) plane



(how this happens)

↳ $\begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -1 & \\ 1 & \end{bmatrix} \rightarrow \begin{bmatrix} -3 \\ 4 \end{bmatrix}$

This is the fn $f(x)$

Application

① Mirroring Images

② Analyzing wave Reflections

Shearing